Term 3 Robotics Challenge 2026 (Control) [20pts]

Trial Runs #2 Checklist: June 2, 2026

Grading Policy & Rules

• Grading is based on the full completion of the 8 control and navigation tasks detailed below.
• Each team is allocated exactly 20 minutes in total (4 minutes for setup/removal and 16 minutes of active testing within their task slot).
• A maximum of 2 trials (1 reset) is allowed per task.
• Penalty: A 1-point deduction will be strictly applied for each manual intervention.

For all tests below apart from the first one, we encourage everyone to program in the Finals' event logic by putting all algorithms within the same code and toggling between them via your mechanical kill switch or otherwise. For instance, if you program your robot to autonomously navigate inside and exit the base, then go through the airlock, enter the arena and start following the grid lines, you can clear tests #2, 3, 5 and 6 in one go.

Checklists

1. Standard Line Tracking [2pts]

Task: The robot must navigate from “Start" (e.g., its deployment area) to the “Destination” (e.g., exit door), following a solid black line smoothly without losing tracking. The robot must successfully finish navigating along a closed circuit within 1 minute. The circuit features smooth turns and no junctions (think a racetrack with mild turns).

- **Success (2 pts):** Successfully navigates the full path within allotted time without losing the line.
- **Partial Credit (1 pt):** Completes at least half of line tracking before a failure occurs or time runs out.
- **Failure (0 pts):** Stops prematurely, departs entirely from the marked track, or fails to advance.

2. Intersection & Tag Alignment [2pts]

Task: The robot must follow a branched line inside the base, correctly identify the first bifurcation, move over the 'Base exit' RFID tag to request exit clearance and continue to the door. Please note that for this test, you can open the door manually if you need, although we require you to try opening it via the provided API.

- **Success (2 pts):** Autonomously follows the correct line branch, stops over the RFID tag and continues to the door within 30 seconds.
- **Partial Credit (1 pt):** Follows the track but misses the branch/tag alignment, requiring manual reset.
- **Failure (0 pts):** Ignores the path intersection entirely, drives off-course, or fails to stop over the RFID tag.

3. Solid Grid Navigation (Left Half Arena) [3pts]

Task: The robot must operate on the left side of the Main Arena, using its sensor array to follow solid grid lines that connect the underlying RFID nodes. Your robot must move forward by two nodes, make a right turn, move by one more node, make a left turn and go forward by another two nodes (a total distance of 1.25 metres, 6 RFID tags; see Fig. 1 below). The manoeuvre must be completed within 1 minute.

- **Success (3 pts):** Smoothly executes grid-line tracking.
- **Partial Credit (2 pts):** Covers over half the distance (until the last turn) or experiences minor tracking oscillations but recovers autonomously.
- **Partial Credit (1 pt):** Covers one third of the distance (the first straight line) or experiences significant tracking oscillations but manages to recover.
- **Failure (0 pts):** Loses tracking on the solid grid lines immediately.

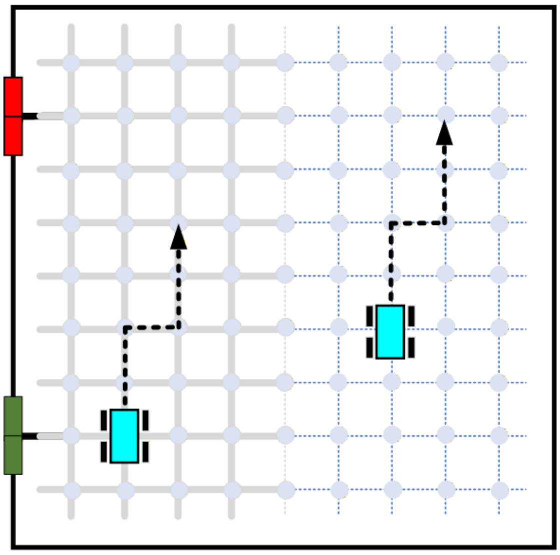

Fig. 1. Manoeuvres required for Tests 3 and 4

4. Open-Field Dead Reckoning (Right Half Arena) [3pts]

Task: The robot must navigate to the right half of the arena, where no solid floor markings exist, moving reliably between spaced RFID hole nodes using dead reckoning, odometry, or localised distance checking. The robot must perform the same manoeuvre as in Task 3.

- **Success (3 pts):** Smoothly executes the manoeuvre within allotted time.
- **Partial Credit (2 pts):** Covers over half the distance (until the last turn) or experiences minor tracking oscillations but recovers autonomously.
- **Partial Credit (1 pt):** Covers one third of the distance (the first straight line) or experiences significant tracking oscillations but manages to recover.
- **Failure (0 pts):** Loses tracking on the solid grid lines immediately.

5. Ramped Incline/Decline Control [2pts]

Task: The robot must successfully ascend the airlock access tunnels, maintaining stable velocity and balance control on the slope. The robot starts within the base, drives through the open doors and the airlock then stops once it is inside the arena.

- **Success (2 pts):** Traverses the entirety of the ramp smoothly without stalling, sliding backward, or tipping over within 30 seconds.
- **Partial Credit (1 pt):** Ascends with severe speed variations or minor slipping but safely completes the ramp transition.
- **Failure (0 pts):** Stalls due to motor limits, tips backward, slips uncontrollably down the incline or gets stalled while transitioning between the base, tunnel or arena.

6. Wall Following [2pts]

Task: Utilising its distance sensors, the robot must smoothly track along the solid walls of the airlocks at a constant distance without experiencing physical collisions.

- **Success (2 pts):** Maintains a consistent, stable distance offset over a 1-meter tracking path within 45 seconds.
- **Partial Credit (1 pt):** Exhibits minor drifting or bumps into the wall surface slightly but manages to self-correct and continue.
- **Failure (0 pts):** Hard crashes repeatedly into the wall boundary or drifts completely away from the sensor tracking threshold.

7. Obstacle Detection and Avoidance [3pts]

Task: The robot must navigate around an obstacle placed within the grid line part of the arena without collisions. The robot starts two nodes away from the obstacle ahead of it and must perform a swerve manoeuvre to return to its original heading once it clears the obstacle and go forward by another 3 nodes. The obstacle only covers a single node. Please see schematic diagram in Fig. 2 for clarifications.

- **Success (3 pts):** Navigates a path around the obstacle safely within 60 seconds.
- **Partial Credit (2 pt):** Suffers minor, transient contact with the obstacles but preserves its trajectory vector or does not return to the straight line immediately after passing the obstacle.
- **Partial Credit (1 pt):** Experiences severe structural scraping or continuous collision against the obstacle or does not return to its original heading after detecting the obstacle.
- **Failure (0 pts):** Directly collides with the obstacle, requiring manual reset.

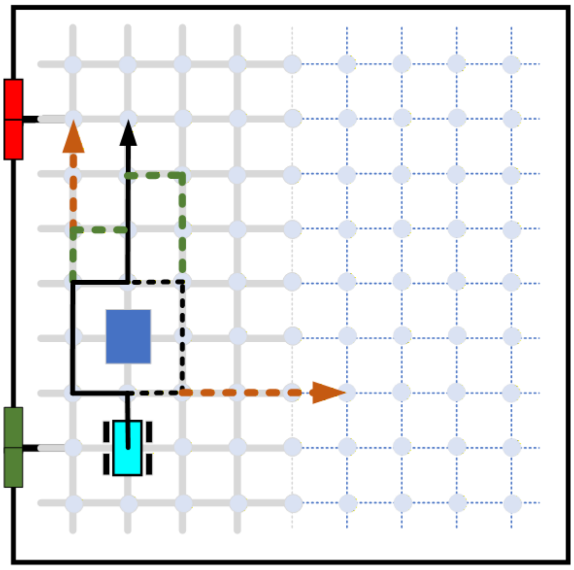

Fig. 2. Swerving around an obstacle for Test 7. Please note that both manoeuvres indicated by black lines (swerving to the left or right) are allowed and will bring you full marks. The manoeuvres indicated by green lines will bring you two points while those shown in orange only a single point.

8. Touch-Based Robot Revival [3pts]

Task: Given the information of a stationary stranded robot, the active robot must execute a controlled deceleration approach and make stable mechanical contact with the target to rescue it. The conditions are recreated as in Task 7, with a stranded robot (an obstacle) placed within 2 nodes away from your robot.

- **Success (3 pts):** Executes a smooth deceleration profile and successful contact the target within 60 seconds.
- **Partial Credit (2 pt):** Successfully triggers the mechanical revival state change but exhibits contact oscillations or pushes the stranded robot forward visibly after the contact.
- **Partial Credit (1 pt):** Successfully triggers the mechanical revival state change but exhibits a high-impact collision overshoot or pushes the stranded robot significantly (off the node).
- **Failure (0 pts):** The robot fails to make contact or glances off the side of the stranded robot.

Performance Log Sheet

Task	Attempt 1	Attempt 2	Interventions (-1pt)	Total Score
1. Standard Line Tracking				/ 2 pts
2. Intersection & Tag Alignment				/ 2 pts
3. Solid Grid Navigation				/ 3 pts
4. Open-Field Dead Reckoning				/ 3 pts
5. Ramped Incline/Decline Control				/ 2 pts
6. Wall Following				/ 2 pts
7. Obstacle Detection and Avoidance				/ 3 pts
8. Touch-Based Robot Revival				/ 3 pts
Total Marks Earned				**/ 20 pts**

Marker Sign-off: ______________________ **Date: June 2, 2026**